Session – VIII: Review of farmer plans and Design Long Term Experiment – November 1st week

Program schedule

S.No	Time	Content	Method	Materials	Expected outcomes
1	20 min.	Recap.	Large group discussion	Charts and markers.	Participants recollect the content that was discussed in seventh session.
2	30 min.	Review of Farmer Decisions.	Small group discussion	Charts and markers.	Participants discuss decisions to be taken for Rabi crops based on available groundwater.
3	30 min.	Conventional crops: Estimation of water requirement & production costs.	Large group discussion and small group discussion.	Charts, markers and calculator.	Participants discuss conventional methods of irrigation and chemical inputs required for a major crop in the HU.
4	30 min.	Alternate crops: Estimation of water requirement & production costs.	Large group discussion and small group discussion.	Charts, markers and calculators.	Participants: • List the major crops and identify more water consuming crops grown in the hydrological unit area. • Discuss the need and advantages of irrigated dry crops.
5	45 min.	Design LTE and selection of collaborator.	Large group discussion	Charts and markers.	Participants select the topic for LTE and collaborator for the conduct of LTE.
6	10 min.	Homework: Crop adoption survey	Large group discussion	Charts and Markers.	Participants discuss ways to collect crop adoption details.
7	10 min.	Evaluation.	Large group discussion	Charts and markers.	Participants share their opinions on the conduct of FWS sessions and methods used in the session.

Learning objectives

By the end of the session participants will be able to:

- Recap the previous session,
- Review Rabi crop plan based on available groundwater balance,
- Discuss conventional crop and estimate water requirement and production cost,
- Discuss alternate crop and estimate water requirement and production cost,
- Discuss the perceived change in terms of water requirement and production cost.

Process

1. Recap Time: 20 min.

Objective : Participants recap the contents discussed in CWB exercise.

Content : CWB workshop discussions

- Objectives of CWB workshop.
- Basic information and area of HU.
- Groundwater recharge.
- Estimation of draft and balance for the end of Kharif.
- Crop plans for Rabi and water requirement.
- Projected recharge through rainfall and secondary water bodies.
- Projected groundwater situation by end of May.

Method : Large group discussion

Materials : Charts and markers.

Process

- Welcome participants and start the session with FWS pledge.
- In large group, encourage participants to recollect the content that was discussed in CWB workshop.
- Use banners or charts (worked by participants during the session) to recollect the content of CWB exercise.

2. Review of farmers decisions Time: 30 min.

Objective : Participants discuss decisions to be taken on Rabi crop plans based on available groundwater.

Content : Change of crop plans.

Method : Small group discussion

Materials : Charts and markers.

Process

- Start the discussion with these words. ”We know the available water balance in the HU and also water requirement for Rabi crops. To meet the water demand or efficient usage of available water, will you rethink about Rabi crop-plans?”
- Request participants to work in small groups and encourage each group to think about their Rabi crop-plans.
- Distribute to each group a handout on Rabi crop plans of their HU or habitation.
- Ask participants to review the crop-plans of their respective HUs, based on the groundwater balance.
- Ask each group to use HESA frame work to present their group work.
- Later, ask each small group to report their work to the large group.
- Following the discussions in the large group, ask each group’s strategy to share their analysis with the other farmers in their habitation.
- Suggest that GMC meetings would be an ideal platform to share this analysis to create awareness among the farmers in the habitation about the groundwater situation. Also, say

that a discussion should be facilitated during the GMC meeting to enable the community/farmers to make concrete decisions on change or revision of crop-plans.

3. Conventional crops: Estimation of water requirement and production cost

Time: 30 min.

Objective : Participants discuss conventional crops and estimate water requirement and production cost.

Content :

- Major crops in the HU and conventional method of irrigation.
- Estimation of water requirement and production cost.

Method : Large group discussion, and small group discussion

Materials : Charts, markers and calculators.

Process

- Ask participants the following question in large group.
    - What are the major crops grown in the hydrological unit?
- List all the major crops on the chart.
- Request participants to share the conventional methods of irrigation that they have been practicing for each identified major crops.
- Write participant responses on the same chart against each major crop.
- Next, request participants to work in small groups to estimate water requirement and production cost for identified crops.
- Encourage each group to present their group work in large group.
- Write the summary of the discussion in the following table.

Conventional crop	Method of irrigation	Water requirement	Production cost

4. Alternate crops: Estimation of water requirement & production cost

Time:30min

Objective : Participants:

- List the major crops and identify more water consuming crops grown in the Hydrological Unit area,
- Discuss the need and advantages of irrigated dry crops.

Content :

- More water consuming crops in the HU.
- Need and advantages of irrigated dry crops.

Method : Large group discussion and small group discussion

Materials : Charts, markers and calculators.

Process

- Say to participants: "We have now discussed conventional crops and water required for those crops. Our Rabi crop plans are also reflecting the same. So we need to rethink about

our crop plans based on groundwater balance. So, let us identify suitable alternate crops that can be grown in the HU area".

- Write participant's responses on a chart paper and request participants to work in small groups and estimate water requirement and production cost.
- Encourage each group to present their group work.
- After group presentations, present the quantity of water required and production cost for conventional crops and alternate crops and discuss the perceived change.
- Encourage participants to take decision on change of crop or alternate method of irrigation to make use of available water.
- Write decisions of participants on their Rabi crop plans on a chart and summarize the discussion.

5. Design LTE and selection of collaborator Time: 45 min**.**

Objective : Participants design LTE and select a collaborator

Content :

- Design a long term experiment.

Method : Large group discussion

Materials : Charts and markers.

Process

Selection of the topic for LTE

- Share the following words with the participants. "Now we are aware of our groundwater situation and water requirement for our crops. If we plan to use available water efficiently we won't get any loss. Let us try to identify what are the major problems in growing crops right from seed sowing to harvesting."
- Divide the participants into two small groups and request the first group to list all the agricultural operations from seed to harvest keeping in view the farmer calendar.
- Request second group to list the problems faced by them during these agricultural operations. Allow this group activity to continue for 15 minutes.
- Ask both the groups to present their group work in large group. Restrict presentation to maximum 10 minutes per group.
- Encourage the participants to arrive at a common understanding with regard to the agricultural operations as well as farmer problems.
- After the group presentation, finalize and list out all the problems on a chart paper.
- Request the participants to share the current practices to overcome these identified problems.
- List all the existing practices on one side of the chart. Then, request the participants to suggest alternate better practices which are listed on the other side of the chart.
- List out all the sources on the same chart paper.
- Request the participants to list the constraints in adaptation of alternate better practices.

- Based on the constraints, encourage the participants to choose the topic that can be experimented in a cost effective way.
- List all these outputs in the table given below and share with the participants.

Problems (or, causes of low yield)	Current practice	Alternate practices / potential for improvement	Constraints	Suggested topics

Tips to facilitator

In case this facilitation takes a long time, based on the situation conduct the following energizer to ease the situation and draw the participant's attention.

Occupation game:

- Divide the participants into five small groups.
- Assign a leader for each group who will act as their spokes person. The group leader will stand facing the members of his group.
- The facilitator will show to the group leaders the name of an occupation. Immediately the group leaders start to enact the occupation shown.
- The first group to name the given occupation will be given a point. The rest of the facilitators may act as the official judges.
- The group which gets the highest number of points will be declared as the winner.

Selection of ideas to be tested

- Continue the discussion after the break, and brainstorm ideas to be tested for an identified topic which can be implemented in the field.
- Ask the participants how they came to know that a particular idea is a better practice and list the source mentioned.
- Encourage the participant to select an idea which can be tested in the field.

Tips to facilitator

Always encourage farmers to consider an idea that does not incur too much expenditure and is devoid of many complications, i.e. experiment should be 'simple'.

- From the list of identified topics, ask the following questions in large group and fill in their collective responses in the table given below.

Put all the outcomes of the discussion in the table mentioned below and summarize the discussion.

Ideas ("*What possible effects will the topic of study have?"*)	Source of each idea	What do we think about each idea? ("*Does it need to be tested?"*)

- Share in the large group that the information in the table would help guide the farmers to plan their observations if the feasible idea is experimented in the field.

Designing the experiment

- Prior to testing the selected idea in the field, we require designing the experiment. To facilitate this discussion, ask the participants in large group, how they plan to perform this experiment in their field. Encourage the participants to implement both their conventional method and the selected experiment at the same time in the field to enable a comparison.
- Ask the participants in large group, how they plan to divide the field for the above mentioned experiment.

Tips to facilitator
Encourage the participants to think about field realities while preparing for LTE. Encourage them to share about a field being uneven with slope, variation in fertility, type of soil, intensity of sun light etc.

- Reach a conclusion on the need for replicating an experiment to overcome natural variation in a farm plot.
- Ask the following question in large group to elicit response from the participants with regard to measures to be taken while having control plot and experiment plot in the same field. List out all the measures on a chart paper.

Farmer plot	**Experiment plot**
Experiment plot	**Farmer plot**

Tips to facilitator

Encourage discussion on the following points:

- There has to be a thorough demarcation between any two plots separated by a narrow bund or trench.
- During taking test plants for observation, always leave the first and the second row to avoid effect of one practice on the other.
- The mentioned effect can be due to fertilizer, irrigation etc.
- Farmer might not agree to waste a piece of land by digging a trench/ bund. In such a case, suggest them to go for intercrop in that area to convince them.

Factors to be observed in an experiment

- In order to analyze the results, it is necessary to observe some important factors in the farmer plot and experiment plot. These observations will provide us with experimental evidence that can be shared with the larger community.
- It will be more useful for the farmer participants to observe the deciding single factor to compare and analyze the results of the experiment.
- Encourage a discussion among the participants on which is the single deciding factor and use the following table to understand the frequency and method of observation.
- List all the responses of the participants in a given table shown below:

WHAT should be observed?	HOW?	WHEN?

6. Homework: Crop adoption survey Time: 10 min.

- Discuss in large group about the need to have crop adoption details. Share that the crop adoption details will help estimate actual draft for Rabi season and the impact of CWB workshop i.e. in increase or decrease in Rabi crop plan shared during the CWB workshop.
- Request participants to collect crop adoption details by involving GMC.

7. Evaluation Time: 10 min.

Ask participants the following questions to evaluate the day's learning. Pose the questions to the large group:

- What are the alternate crops suitable to your HU area?
- List effective alternate methods of irrigation.

Thank the participants for their collective decisions on Rabi crop-plans.